24/11/20

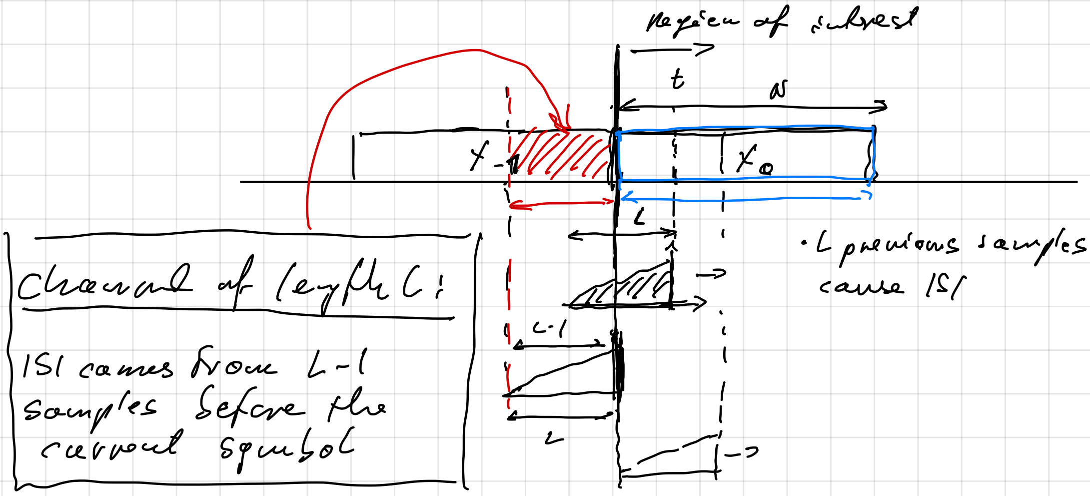
$\Rightarrow$ Only first L samples are affected by ISI

$$\bar{y}[t] = \sum_{\tau=0}^{L-1} h[\tau]\,\bar{x}[t-\tau]$$

consider only $0 \le t \le L-1$ corresponds to receiving x_0

$\Rightarrow$ which samples affect the first L samples $x_{-1}[N-(\tau-t)]$

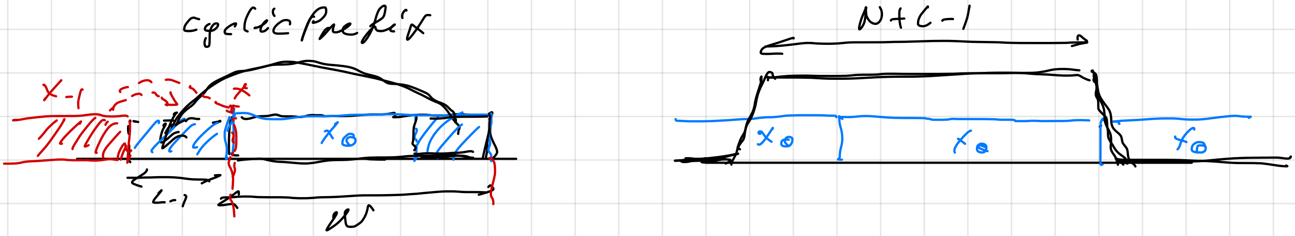
$$\bar{y}[t] = \sum_{\tau=0}^{\min(t,L-1)} h[\tau]\,\underbrace{\bar{x}[t-\tau]}_{x_0[t-\tau]} + \underbrace{\sum_{\tau=t+1}^{L-1} h[\tau]\,\overbrace{\bar{x}[t-\tau]}}_{\text{Disappears for } t \ge L}$$



OR $\boxed{\bar{y}[t] \quad 0 \le t < N \quad \text{depends only on } \bar{x}[t] \text{ with } -L+1 \le t < N}$

Trick to remove ISI for $0 \le t < N$ the "Relevant samples"

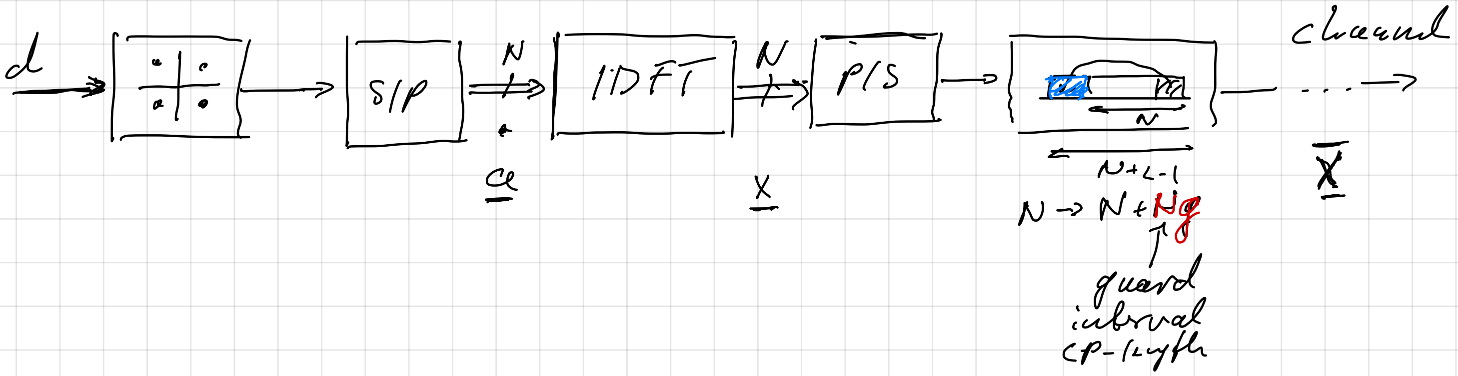
@TX must correspond to the periodic signal $x_K[t]$

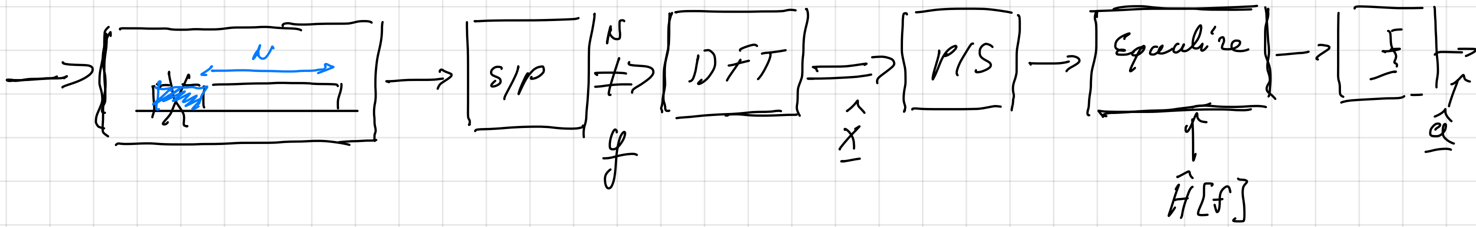


1) Use a periodic signal with a window of $N+L-1$

2) Start from a limited time signal $0 \ldots N-1$ and prepend a cyclic prefix

OFDM Transmitter + Receiver





$$\underline{a} = [a_0, a_1, \dots, a_{N-1}]^T$$

$$\underline{x} = [x_0, x_1, \dots \underbrace{x_{N-1}}_{CP}]^T \xrightarrow{CP} \bar{\underline{x}} = [\underbrace{x_{N-N_g}, x_{N-N_g+1}, \dots, x_{N-1}}_{CP}, x_0, x_1, \dots x_{N-1}]$$

$$\underline{h} = [h_0, h_1, \dots, h_{L-1}] \qquad \underline{y} = [y_0, y_1, \dots, y_{N-1}]$$

1) Write IDFT in matrix notation

DFT matrix $\underline{\underline{F}}$: $f_{k,\ell} = e^{j\frac{2\pi}{N}k\cdot\ell}$

$$w = e^{j\frac{2\pi}{N}}$$

$$\underline{\underline{F}} = \begin{bmatrix} 1 & 1 & \dots & 1 \\ w^0 & w^1 & & w^N \\ & & \vdots & \\ w^0 & w^{N-1} & \dots & w^{(N-1)^2} \end{bmatrix}$$

$k \downarrow$, $\overrightarrow{\ell}$, $w^{k\cdot\ell}$

$$\underline{X} = \underline{\underline{F}}^{-1}\underline{a} = IDFT\{a\}$$

2) Add the CP as shown above

3) Express linear convolution of $\bar{\underline{x}}$ with the channel to get the received signal

$\underline{h} = [h_0, h_1, \cdots h_{L-1}]$ $\quad L-1 = N_g$ (channel has as many taps as N_g CP)

1st sample

$$\begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_{N-1} \end{bmatrix} = \underbrace{\begin{bmatrix} h_{N_g} & h_{N_g-1} & \cdots & h_0 & 0 & \cdots & 0 \\ 0 & h_{N_g} & \cdots & h_1 & h_0 & 0 \cdots & 0 \\ & & & & & & \\ & & & & h_{N_g} & \cdots & h_0 \end{bmatrix}}_{\text{Toeplitz Matrix}} \underbrace{\begin{bmatrix} x_{N-N_g} \\ x_{N-N_g+1} \\ \vdots \\ x_{N-1} \\ x_0 \\ \vdots \\ x_{N-1} \end{bmatrix}}_{\underline{\bar{x}}}$$

CP

identical

We use that the first and last entries of $\underline{\bar{x}}$ are the same to form the Toeplitz matrix into a Circulant Matrix

$$N \times (N + \overset{N_g}{L} - 1) \rightarrow N \times N$$

$$\begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_N \end{bmatrix} = \underbrace{\begin{bmatrix} h_0 & 0 & \cdots & 0 & h_{N_g} & h_{N_g-1} & \cdots & h_1 \\ h_1 & h_0 & 0 & \cdots & 0 & h_{N_g} & h_{N_g-1} \cdots & h_2 \\ & & & & & & & \\ 0 & \cdots & & \cdots & 0 & h_{N_g} & \cdots & h_0 \end{bmatrix}}_{\text{Circulant Matrix } \underline{\bar{H}} \quad (N \times N)} \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{N-1} \end{bmatrix}$$

For circulant Matrices $\underline{\bar{H}} = \underline{F}^{-1} \underline{\Lambda} \underline{F}$

- SVD: Columns of DFT matrix (complex sinusoids) are Eigenvectors

$\underline{\Lambda} = \text{diag}\{DFT\{h\}\}$

4) Substitute SVD into (Tx + Channel) $\quad \underline{y} = \overset{H\bar{x}}{\underline{H}\,\underline{x}} = \underline{F}^{-1} \underline{\Lambda} \underbrace{\underline{F}\,\underline{F}^{-1}}_{I} \underbrace{\underline{F}^{-1} \underline{a}}_{x} = \underline{F}^{-1} \underline{\Lambda}\, \underline{a}$

5) <u>Receiver:</u> DFT of $\underline{y}$: $\hat{x} = \underline{F}\,\underline{y} = \underbrace{\underline{F}\,\underline{F}^{-1}}_{I} \underline{\Lambda}\, \underline{a} = \underline{\Lambda}\, \underline{a} = \begin{bmatrix} H[0]\, a_0 \\ H[1]\, a_1 \\ \vdots \\ H[N-1]\, a_{N-1} \end{bmatrix}$

Equalizer: $\hat{\underline{a}} = \underline{\underline{\Lambda}}^{-1} \underline{\underline{F}}\, \underline{a}$

$$\hat{a}_K = DFT\{h\}_K^{-1} \cdot DFT\{y\}_K$$

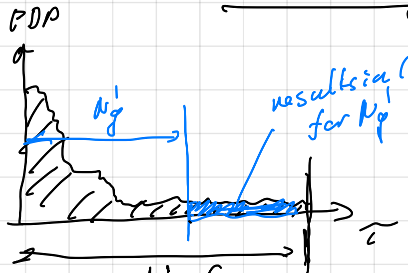
System design considerations

1) Choosing the Cp length?

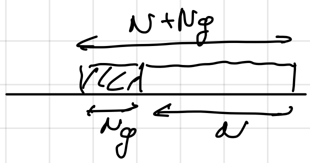
- $N_g > L$: greater than the channel length
  → related to the length of the channel PDP
  a) CP too short: I have ISI
  b) CP to long: ??

Efficiency:

- OFDM Symbol duration: $N+N_g$
- Symbols in an OFDM symbol is N



$$\underline{\underline{\mathcal{E} = \frac{N}{N_g+N}}}$$

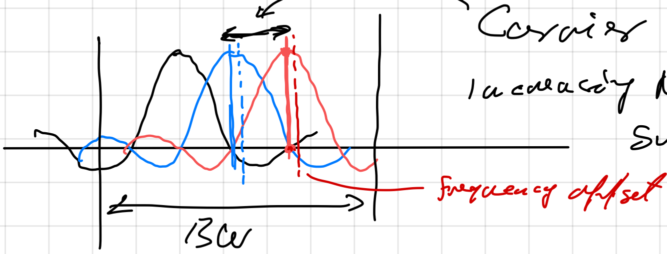


For a given N, increasing N_g reduces the efficiency

2) How to choose N?

- Efficiency: $\mathcal{E} = \frac{N}{N_g+N}$
- Consider $N \to \infty \Rightarrow \lim_{N\to\infty} \frac{N}{N_g+N} \to 1$ ✓

However!



Carrier spacing: $\frac{2\pi}{N} \cdot \frac{1}{T}$

Increasing N squeezes the subcarriers closer

More subcarriers → more sensitivity to frequency offsets